Analysis of Salary Data

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Agenda

Findings of linear regression modeling with tech salary data

- Data Description
- Regression Results
- Interpretation and Next Steps

Data Description

Dataset Overview

- Dataset contains 373 employee records after removing missing values
- Features include:
- Age
- Gender
- Education Level
- Job Title
- Years of Experience
- Salary

Missing Data

- The dataset contained 2 missing values in several columns
- Missing observations were removed using dropna()

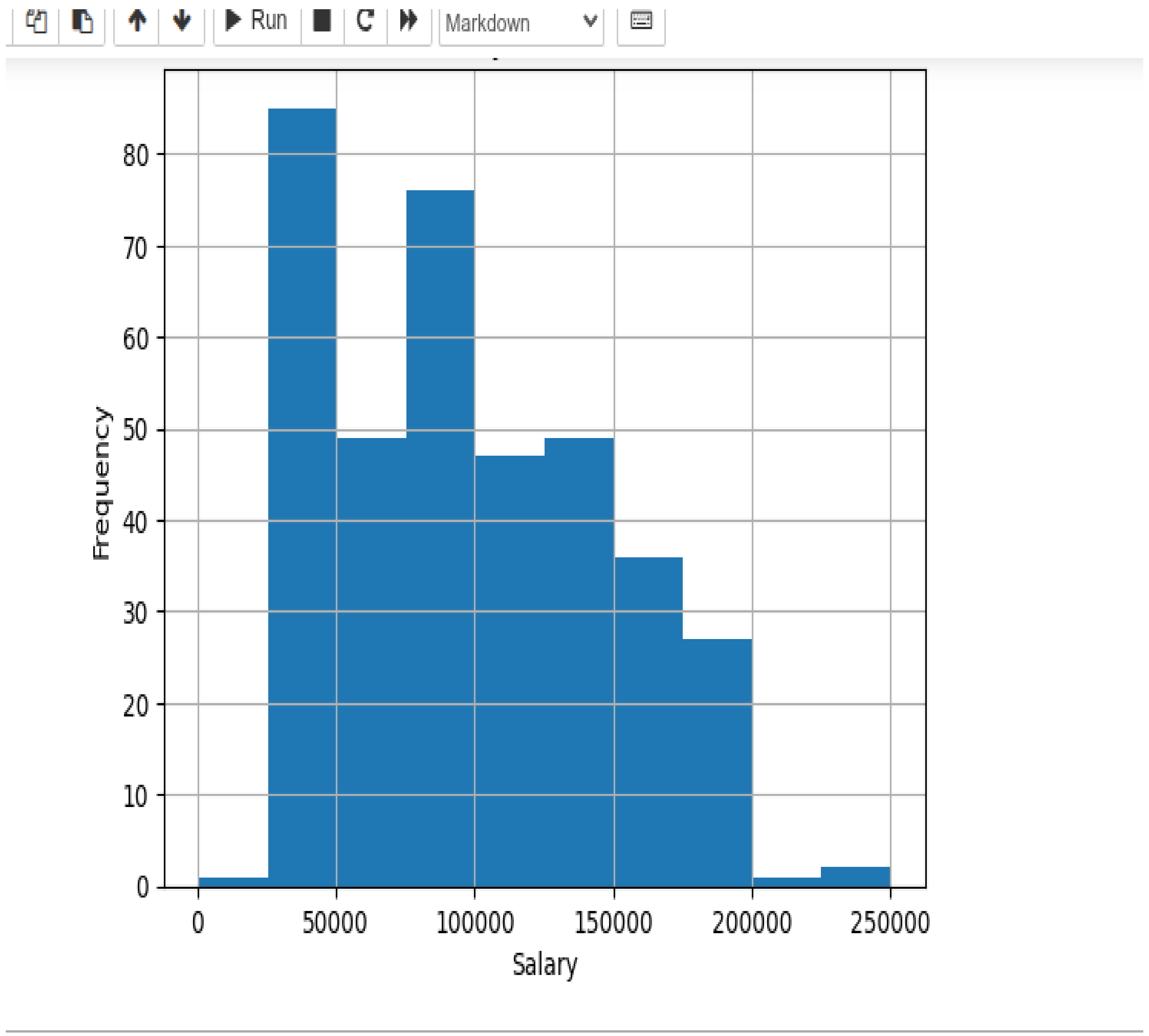
Salary Range

- Minimum salary: $350
- Maximum salary: $250,000
- Average salary: approximately $100,577

Salary Distribution

Key Findings

- Salary distribution is positively skewed
- Most salaries fall between approximately $30,000 and $150,000
- A small number of high-income observations create a long right tail



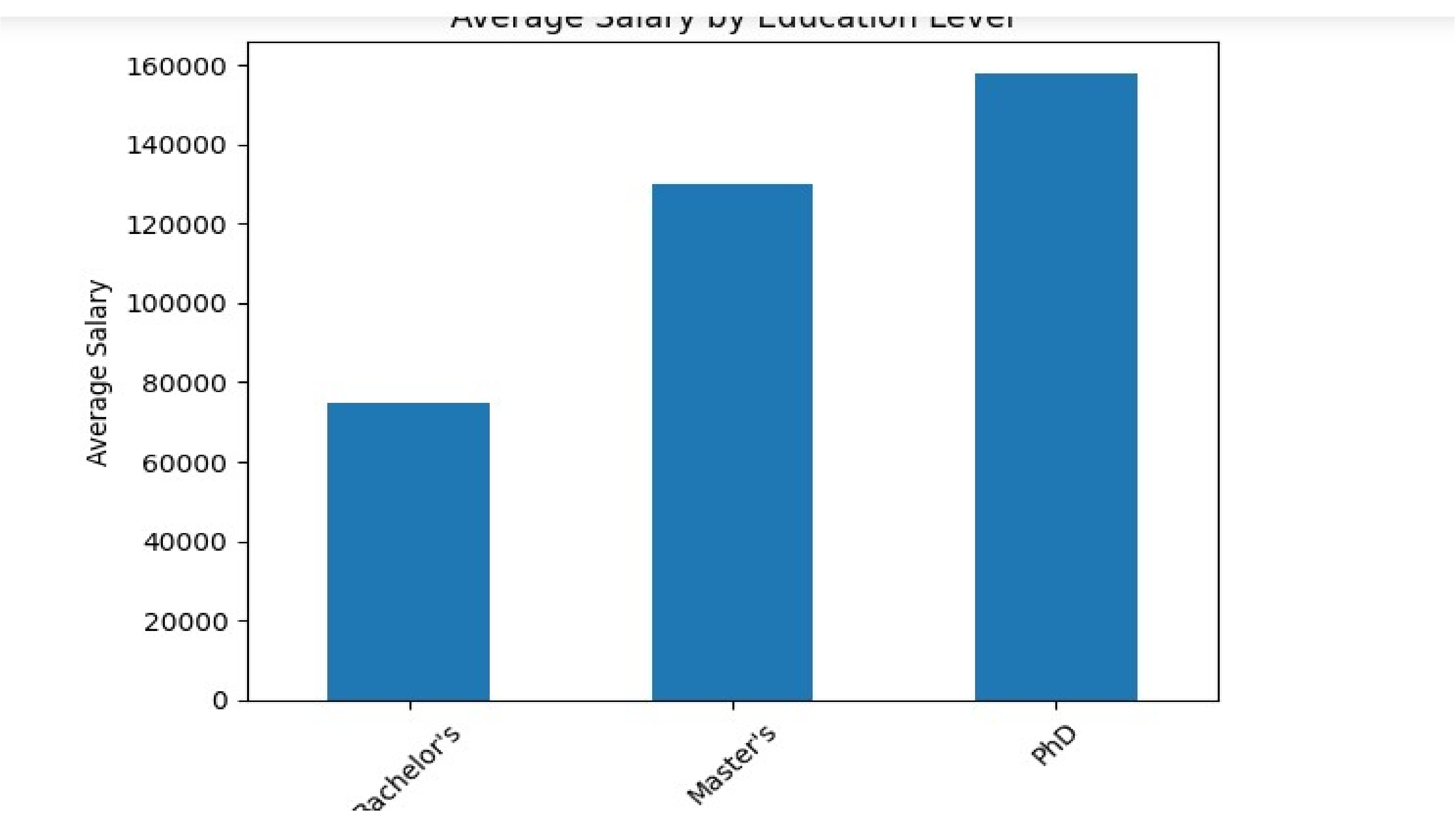
Education Level Analysis

Average Salary by Education Level

Education Level	Average Salary
Bachelor's	$74,756
Master's	$129,796
PhD	$157,843

Observation

- Higher education levels are associated with higher salaries.



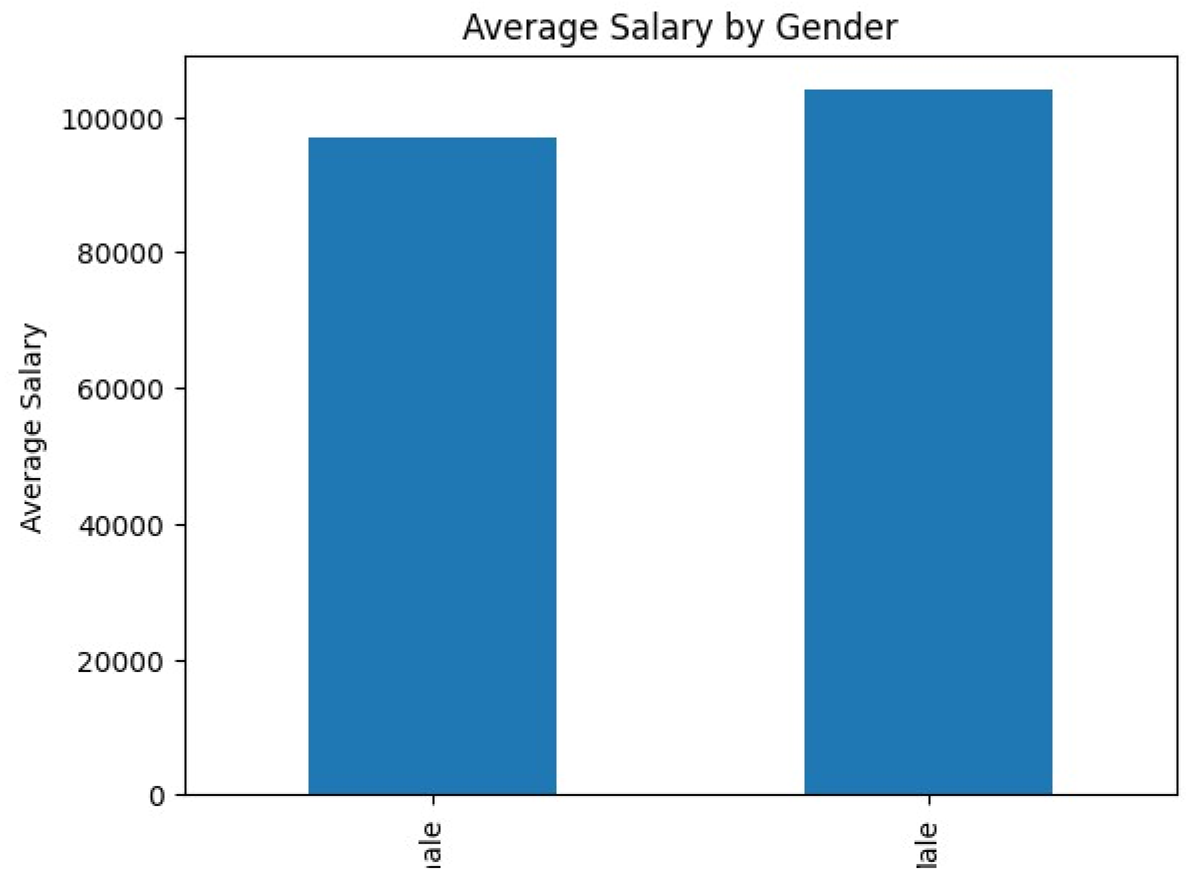
Gender Analysis

Average Salary by Gender

Gender	*Average Salary*
Female	$97,011
Male	$103,868

Observation

Male employees earned a higher average salary in this dataset.



Experience Analysis

Salary vs Years of Experience

Observation

- Salary generally increases with years of experience
- Relationship appears strongly positive and approximately linear

Regression Results

Initial Regression Model

Initial Model Included

- Years of Experience
- Gender
- Education Variables
- Job Title Indicators

Non-Significant Variables Removed

- is_junior
- is_analyst
- is_engineer
- because their p-values were greater than 0.05.

Regression Results

Final Regression Model

Model Fit

- $R^2 = 0.914$
- Model explains approximately 91.4% of salary variation

Final Significant Variables

- Years of Experience
- is_male
- is_director
- is_senior
- is_manager
- Education Level_Master's
- Education Level_PhD

Interpretation and Next Steps

Interpretation — Experience

Years of Experience

Each additional year of experience increases salary by approximately $5,104

95% Confidence Interval:

$4,756 to $5,453

Interpretation — Education

PhD Effect

Employees with a PhD earn approximately $23,080 more than employees without a Master's or PhD degree

95% Confidence Interval:

$17,700 to $28,500

Interpretation and Next Steps

Predicted Salary Example

For:

- male employee
- 5 years of experience
- senior title
- bachelor's degree

Predicted salary:

- $76,362

Actual salary example:

- $110,000

Residual:

- $33,638

Conclusions

- Years of experience is the strongest predictor of salary
- Higher education levels are associated with substantially higher salaries
- Senior and director titles significantly increase salary
- The regression model provides a strong fit with R^2 of 0.914

Potential Improvements

- Include additional variables such as industry, location, or company size
- Investigate outliers and residual distributions
- Test nonlinear relationships and interaction effects